

BoilerTronics User Guide : Game Overview and Explanations

General Mechanics Overview

December 4, 2025

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1 Overview

This document highlights important elements within each interface of the game including the main menu, level interface, and level creator. Details regarding each selectable option are explained along with images to illustrate the location of each interactable component to serve as an overall guide to the user to clarify necessary information before and during gameplay.

2 Main Menu

Upon running the game, the first component of the game you can interact with is the Main Menu. From the main menu, you have many options such as being able to start a new game, select a level, and adjust your settings as the primary three options.

You can also choose the icons among the bottom row which allow you to view your profile, view the leaderboard, access the level creator (for developers), or exit the game.

2.1 Menu Options

The main three options the user can interact with upon opening the game is:

- New Game - Create a new game, unique from previous saved slots where one can “start fresh”
- Level Select - Choose a level (within a section) to play by scrolling to the left and right to move the train and enter the level by clicking play, otherwise choosing back to return to the main menu
- Level Creator - Used for creation and development of levels, described in Section 4 of the User Guide



2.2 Alternate Menu Options

Additionally, the menu contains four buttons below the primary options displayed as icons. These four options (from left to right) are:

- Profile - See your profile and friends, along with earnable/earned achievements and easter egg menus
 - Profile login/registration - You can enter your email and password and select the login option if you have already registered or login for an existing profile

- Friends - The tab on the left which will show your existing friends list and allow for the addition of new friends
 - Enter email and password on the left panel to be able to login/register to see friends and your profile options



- Friends tab on the right allows for the addition of friends via the username lookup as well as the ability to accept/decline incoming friend requests and a list of current friends
- Logout option allows you to sign out of your account to display the ability to log back in or register an account



- Achievements - locked and unlocked achievements which highlight the requirements to obtain them and update upon completion
- Easter Eggs - locked and unlocked easter eggs which hint on how to obtain them and update upon completion
- Leaderboard - See your score and the top six scores of your friends by level
- Settings - Make changes to the in-game volume, adjust fullscreen status, and open the User Guide or Manual
- Exit - Exit the game

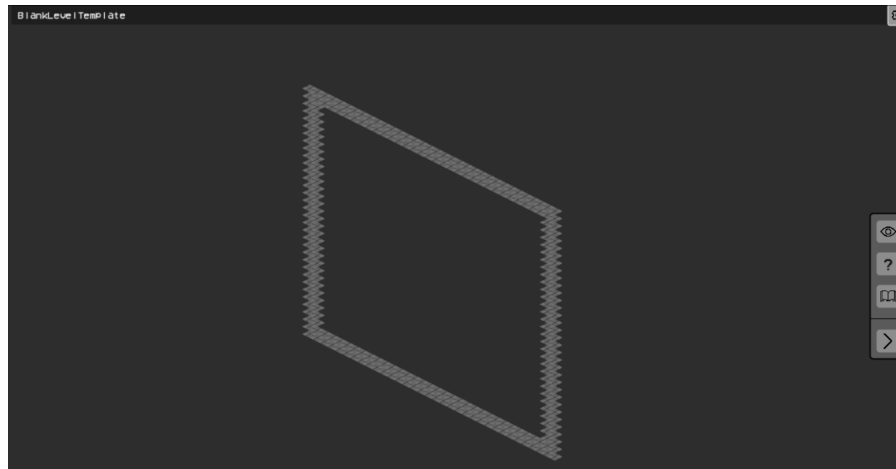


3 Level Interface

The 'level interface' defines the user interface of the level as the sum of the components you interact with to contribute towards constructing the solution of the level as well as related features and options.

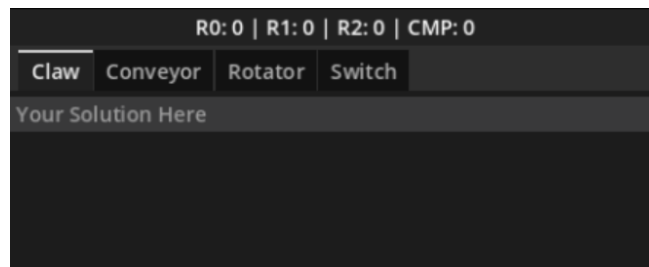
3.1 Level "Field"

The level field (blank example shown below) is the area in which objects can be placed to complete a level solution. The staggered border defines the limits of where objects can be placed and are not included as being "in-bounds".



3.2 Terminal(s)

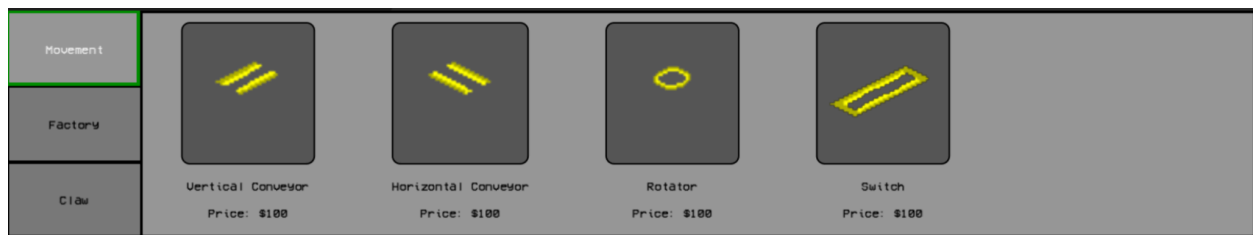
The terminals for all programmable objects placed onto the level will have a corresponding terminal in the top left of the level. Object code is typed below its corresponding objects with each tab being selectable and orderable to allow for customability in its functionality. When solutions are run, the current line of execution is highlighted white while errors cease program execution and are highlighted red.



3.3 Layers

The objects which can be placed onto the level “field” are divided into layers based on their function and related interactability. To switch between layers for item removal and addition, you must click the corresponding layer in the bottom left to interact with objects belonging to that layer. There are three layers facing the player-side with an additional for developers as listed below:

- Movement Layer
 - Conveyors - Have a vertical and horizontal variation which are used to move rails and claws up and/or down for the vertical variant and left and/or right for the horizontal variant
 - Rotators - Rotates the rail on top, the same rail can therefore be used horizontally and vertically
 - Switch - Items placed on either tile of the switch swap positions



- Factory Layer
 - Furnace - “Cooks” the item placed inside, also needs fuel to do so
 - Roller - Rolls out the item placed inside, creates a variation of the item inserted
 - Press - Presses the item placed inside, creates a variation of the item inserted
 - Roller - The roller allows you to turn a sheet of material into a pipe of that given material though no fuel is needed
 - Press - The press allows you to turn a bar of a given material into a sheet of that given material though no fuel is needed
 - Pipe - Used to transport water/lubricant from the pump
 - Pump - Source of water/lubricant to be moved using pipes



- Claw Layer
 - Claws - must be placed on a rail to function, necessary for the completion of any level as it is used to grab and drop items
 - Rails - Have a vertical and horizontal variation which dictates the direction in which a claw can be moved



- Developer Layer
 - Input objects - Green boxes which claws can grab input items from to interact with for level completion
 - Output objects - Blue boxes which claws can drop the corresponding item into for level completion



3.4 Time Controls

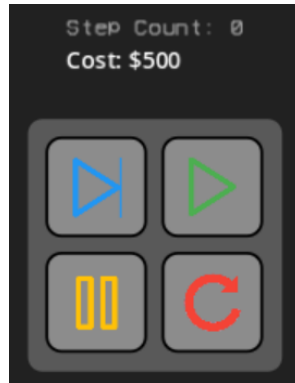
The bottom right section includes live updating statistics as well as four buttons used to control the flow of the solution.

Live Statistics:

- Step Count - live tracks the number of steps taken for the current running solution starting from 0
- Cost: dollar value of current solution (whether complete or not) and live updates based on objects added to the level as well as when objects are removed

Time Control Buttons:

- Step Button (Blue) - Used to iterate one time step through the solution
- Play Button (Green) - Used to step through the level automatically, subsequent clicks will increase the stepping speed of the level
- Pause Button (Yellow) - Used to pause the current running solution after the play button
- Reset Button (Red) - Reset the state of the level, moving objects back to their original location and stopping the actively running solution



3.5 Accessibility Options

The right side collapsible menu is used for general accessibility when it comes to gameplay with its three options.

- Layer visibility - The top option will allow you to toggle the visibility of each layer within the level interface to more clearly see each one
- Hints - Opens the hints popup which contains a series of scrollable hints by clicking the next and previous arrows
- Documentation - Opens the reference manual containing information for terminal programming including relevant information on arithmetic, registers, control flow, and instructions
- Collapse - the bottom, separated button allows for the accessibility menu to be collapsed to a small rectangle which can be clicked to reopen the menu

3.6 Settings

The settings gear icon in the top right can be selected to open up a general settings menu with four selectable options.

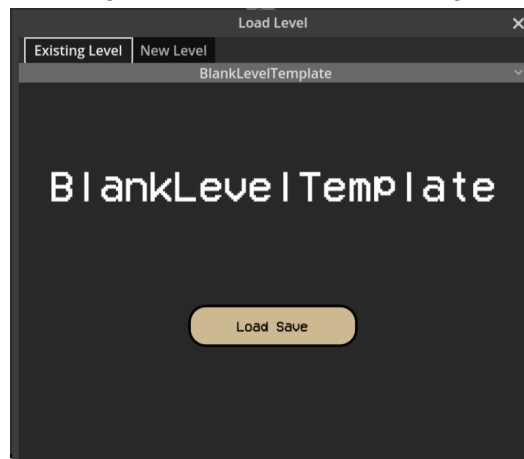
- Save - Save the current level solution whether complete or incomplete to one of three save slots which are red if unused and green if already storing level information. These save slots can be overridden via the clear button underneath to clear the information within the save slot to be able to use
- Statistics - Shows the statistics for the current, completed level solution across the three measured statistical categories including production per step, cost, and steps. The menu will initially be set to default values for the cutoffs but updates the statistics once a solution is reached, highlighting 'good' performance in green and 'poor' in red
 - Enhanced Statistics -
- Settings - Similar to the main menu settings, includes adjustable volume/mute settings along with the ability to toggle fullscreen for the game
- Exit - Exits the current level to return to the main menu

4 Level Creator

The level creator defines the development tool which allows for levels to be created and saved to the local file system to be able to add to the Godot file system to be able to load. These created levels can also be shared with others to be able to load and play.

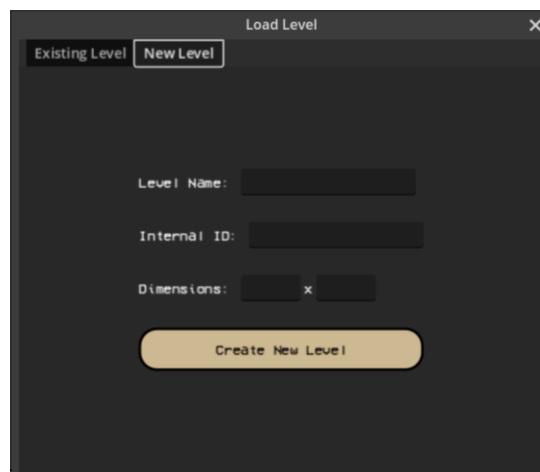
4.1.1 Load Existing Level

The initial menu of the level creator includes the ability to load an existing level by selecting the level name in the dropdown, clicking on load save, then closing the load level menu



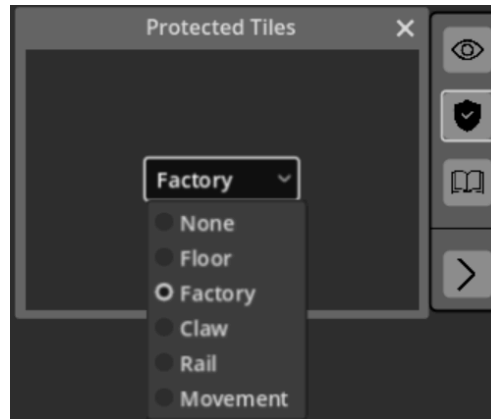
4.1.2 Load New Level

The initial menu of the level creator also includes the ability to load a new level by entering a name, internal id, dimensions for the level, and clicking to create a new level before closing the load level menu



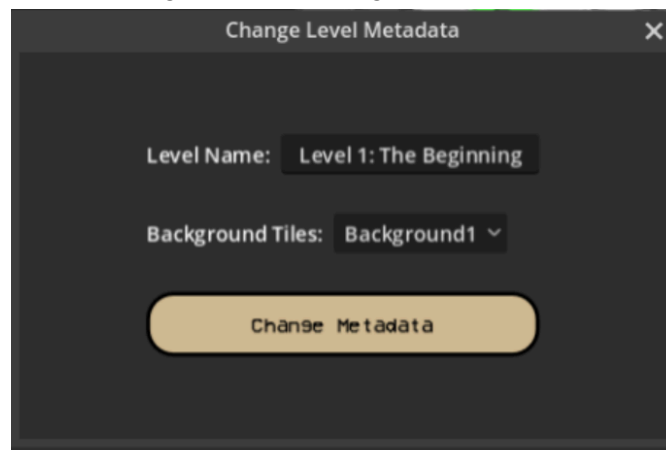
4.2 Protected Layer Objects

The level creator also allows for protecting objects in certain layers to prevent them from being moved or altered in any way regarding its position. After selecting the protection option and selecting the layer to change the protection status of, you can see protected objects as being highlighted blue. This can be altered by hovering over the item and pressing the up arrow on the keyboard to toggle protection on and off.



4.3 Change Level Metadata

Select the pencil icon in the top right next to the setting icon to change the level metadata including the level name and background, selecting the button at the bottom to confirm changes.



4.4 Save Created Level

To save the level to your local file system, select the settings gear icon in the top right then select the save option. Enter the name of the file to save to your local file system and click export to finalize the save as the file path for the save is noted above the button.

5 Notes for Use

Some additional notes for use of the game with clarifications on game mechanics are noted below.

5.1 Placing Objects onto Level

Objects placed onto the level will automatically 'snap' into place due to the use of a grid system. Additionally, items are placed dependent on the location of the mouse, **not** the location of where the image of the object appears. Objects can also be deleted from the level by hovering over the object and right clicking.

In addition to placing objects, factory objects can be rotated on the layer view at the button by right clicking which will also reflect the appearance of the object when placed onto the level.

When placing input and output objects for a claw to interact with, ensure that it is placed 'below' the claw such that there is a vertical space separating the two (should not be stacked directly on one another on the same tile).

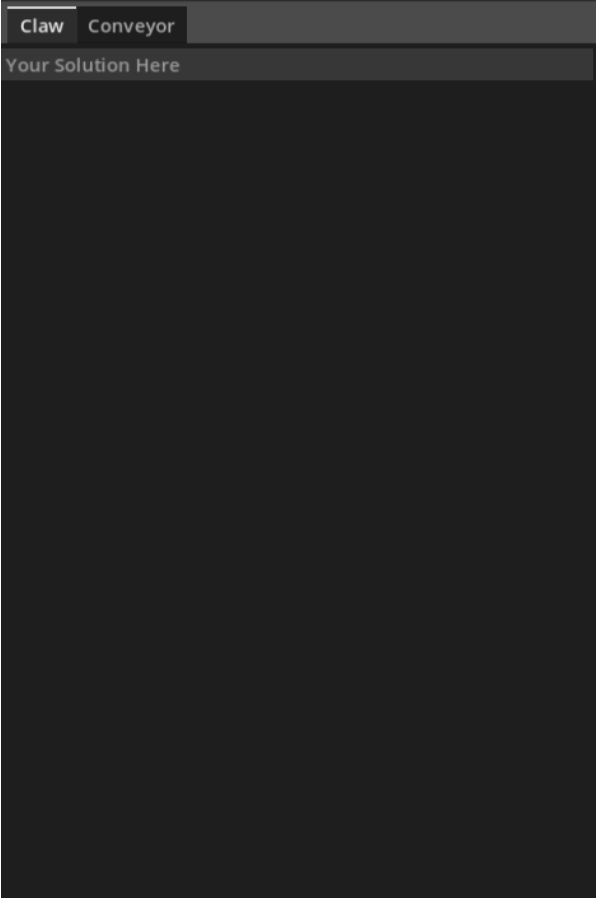
5.2 Zooming & Panning

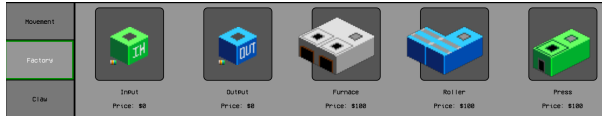
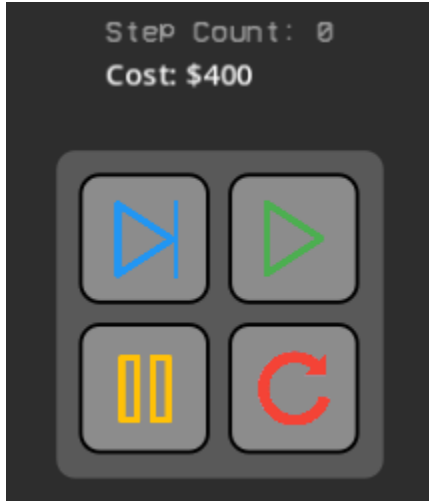
The level interface can have its appearance altered by zooming in using the scroll wheel of the mouse. Additionally, you can pan across the level by holding the right click while clicking and dragging using the left mouse button to shift the level in any direction.

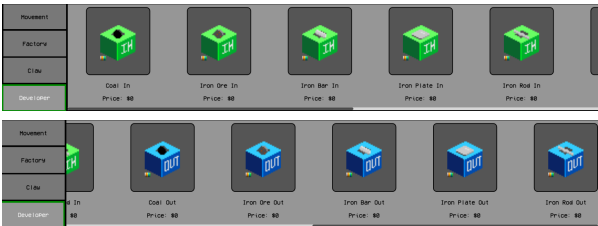
5.3 Tooltips

When hovering over items in the item picker at the bottom where the layers can be selected, tooltips will open. Tooltips will vary based on the item which you are hovering over and will include general information regarding it. The tooltips will close once the mouse is moved back to the level or replaced if the mouse is moved over another item.

OLD STUFF BELOW

Main menu icon buttons Left to right: • Profile - user profile and user information • Leaderboard - leaderboard of other and personal level stats • Level creator/documentation - opens level creator • Exit - exits game	
Terminals • Tabs across the top for selecting which object to code • Code goes in “textbox” underneath • See Assembly manual for details on the programming	
Layers • Select the layer on the left • To place object on the field click the objects to the right of the 3 tabs then left-click on the field to place • To remove an object from the field, right-click the object • NOTE: the location of the mouse is where the object will be placed,	

NOT where the image of the object is	
Layer Objects • Movement Layer ○ For moving/rotation of rails • ClawLayer ○ Includes claws and rails • Factory Layer ○ Input/Output - resource collection and output ○ Furnace/Roller/Press - modifies resources for output specification	Movement Layer:  Claw Layer:  Factory Layer: 
Buttons • Top 2 texts are the step count and cost which update when necessary • Blue top left ○ Step button, advances solution by 1 step • Green top right ○ Run button, steps automatically • Yellow bottom left ○ Pause button, pauses running of solution • Red bottom right ○ Reset button resets solution to original state (pre stepping/running)	

Button Menu From top to bottom: • Visibility - toggle visibility for each layer • Hints - Gives the player hints regarding the level/game • Documentation - To be implemented, sprint 3 • Close button - closes the button menu (Can be reopened using the small rectangular button which pops up after closing) Note: the developer side has a protection button which prevents a player from changing specified objects 	
Settings • At top right of level ui	
Zooming + Panning • Use the scroll wheel to zoom (zooms in/out based on the center) • Pan across the level by right-clicking and holding then moving the mouse	No image
Developer Layer: The level creator (developer-side) has an extra layer with additional input/output factory elements Note: There are more elements in the dev later than shown on the left	

Load Level popup

Opens when the user initially opens the level creator allowing for loading of an existing level or creation of a new level

